

Simply Connected Regions

More about the validity of Green's theorem.

We have seen Green's theorem

$$\oint_C \vec{F} \cdot \hat{T} ds = \iint_R \text{curl } \vec{F} dA$$

or

$$\oint_C \vec{F} \cdot \hat{n} ds = \iint_R \text{div } \vec{F} dA$$

This only works if $\vec{F}$ (and derivatives) are defined everywhere in R .

Example

$$\vec{F} = \frac{-y\hat{i} + x\hat{j}}{x^2 + y^2}$$

$\vec{F}$ is not defined at the origin.

$$\text{curl } \vec{F} = 0$$

everywhere else.

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl } \vec{F} dA = 0$$

We cannot use Green's theorem directly.

$$\oint_C \vec{F} \cdot d\vec{r} - \oint_{C_0} \vec{F} \cdot d\vec{r} = \iint_R \text{curl } \vec{F} dA$$

Total Line Integral

$$\oint_{C'} - \oint_{C''}$$

(the white segments cancel out)

Note

Definition: A connected region in the plane is simply connected if the interior of any closed curve in R is also contained in R .

If the domain where $\vec{F}$ is defined and differentiable is simply connected, then we can always apply Green's theorem.

If

$$\text{curl } \vec{F} = 0$$

and the domain where $\vec{F}$ is defined is simply connected, then $\vec{F}$ is conservative (a gradient field).

Example

$$\int_0^1 \int_x^{2x} dy \, dx$$

Exchange the order:

$$\int_0^{1/2} \int_0^y f \, dx \, dy + \int_{1/2}^1 \int_y^1 f \, dx \, dy$$

Polar Coordinates

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx \, dy = r \, dr \, d\theta$$

Remember:

$$\text{Area}(R) = \iint_R 1 \, dA$$

Mass, average value $\bar{f}$, centroid $(\bar{x}, \bar{y})$.

Polar moment of inertia:

$$I_0 = \iint (x^2 + y^2) \rho \, dA$$

$$I_0 = I_x + I_y$$

Useful Integration Methods

- Easy trigonometric identities
- Substitution
- Hyperbolic trigonometric substitution
- Integration by parts

Example:

$$u = 1 + t^2$$

$$\int \frac{t \, dt}{\sqrt{1 + t^2}}$$

Change of Variables

$$u = u(x, y), \quad v = v(x, y)$$

Jacobian:

$$\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$$
$$du \, dv = \left| \frac{\partial(u, v)}{\partial(x, y)} \right| dx \, dy$$

Substitute x, y in the integral.

Setting Up Bounds

Line integral:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C M \, dx + N \, dy$$

where

$$\vec{F} = \langle M, N \rangle$$

Flux form:

$$\int_C \vec{F} \cdot \hat{n} \, ds = \int_C -N \, dx + M \, dy$$

Evaluation

Reduce to a single parameter:

$$x = x(t), \quad y = y(t)$$

Express everything in terms of t and integrate with respect to dt .

Conservative Vector Field

If

$$\text{curl}(\vec{F}) = N_x - M_y = 0$$

and the domain is simply connected, then $\vec{F}$ is a gradient field.

$$f_x = M, \quad f_y = N$$

$$\Longleftrightarrow \quad \vec{F} = \nabla f$$

Then

$$\int_C \nabla f \cdot d\vec{r} = f(P_b) - f(P_a)$$

Green's Theorem

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl } \vec{F} \, dA$$

where

$$\text{curl } \vec{F} = N_x - M_y$$

Also,

$$\oint_C \vec{F} \cdot \hat{n} \, ds = \iint_R \text{div } \vec{F} \, dA$$

where

$$\text{div } \vec{F} = M_x + N_y$$

Lecture 26: Triple Integrals

$$\iiint_R f \, dV$$

where the volume element is

$$dV = dx \, dy \, dz$$

Example 1

Region between paraboloids:

$$z = x^2 + y^2$$

$$z = 4 - x^2 - y^2$$

Volume:

$$\iiint_R 1 \, dV$$

Set up the integral:

$$\int_{-2}^2 \int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}} \int_{x^2+y^2}^{4-x^2-y^2} dz \, dy \, dx$$

Find the shadow in the xy -plane where

$$z_{\text{bottom}} < z_{\text{top}}$$

So,

$$x^2 + y^2 < 4 - x^2 - y^2$$

$$2x^2 + 2y^2 < 4$$

$$x^2 + y^2 < 2$$

Therefore, the projection is a disk of radius

$$\sqrt{2}$$

Using Polar Coordinates

It is better to use polar coordinates instead of x, y .

$$\int_0^{2\pi} \int_0^{\sqrt{2}} \int_{r^2}^{4-r^2} dz \, r \, dr \, d\theta$$

This is easier to evaluate.

Cylindrical Coordinates

$$(r, \theta, z)$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

Volume element:

$$dx \, dy \, dz = r \, dr \, d\theta \, dz$$

Also,

$$\Delta V = \Delta A \cdot \Delta z$$

$$dV = dA \cdot dz$$

Application: Mass Density

Density:

$$\delta = \frac{\Delta m}{\Delta V}$$

Hence,

$$dm = \delta dV$$

Mass of the solid:

$$\iiint_R \delta dV$$

Center of Mass

$$(\bar{x}, \bar{y}, \bar{z})$$

$$\bar{x} = \frac{1}{\text{mass}} \iiint_R x \delta dV$$

In this example,

$$\bar{x} = \bar{y} = 0$$

by symmetry.

Moment of Inertia

Moment of inertia about an axis:

$$\iiint_R (\text{distance to axis})^2 \delta dV$$

For the z -axis:

$$I_z = \iiint_R r^2 \delta dV = \iiint_R (x^2 + y^2) \delta dV$$

For the x -axis:

$$I_x = \iiint_R (y^2 + z^2) \delta dV$$

For the y -axis:

$$I_y = \iiint_R (x^2 + z^2) \delta dV$$

Example 2

I_z of a solid cone between

$$z = ar$$

and

$$z = b$$

with density

$$\delta = 1$$

Integral setup:

$$I_z = \iiint r^2 \cdot r \, dr \, d\theta \, dz$$

Equation of cone:

$$z = ar \quad \Rightarrow \quad r = \frac{z}{a}$$

Moment of Inertia of a Cone

$$I_z = \int_0^b \int_0^{2\pi} \int_0^{z/a} r^2 \cdot r \, dr \, d\theta \, dz$$

Evaluating gives:

$$I_z = \frac{\pi b^5}{10a^4}$$

Example 3

Set up

$$\iiint_R dV$$

for the region

$$z > 1 - y$$

inside the unit ball centered at the origin:

$$x^2 + y^2 + z^2 = 1$$

Integral setup:

$$\int \int \int dz \, dx \, dy$$

with bounds

$$1 - y \leq z \leq \sqrt{1 - x^2 - y^2}$$

Projection onto the xy -plane satisfies:

$$1 - y \leq \sqrt{1 - x^2 - y^2}$$